# Strength of materials
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## IDENTIFICATION

CODE : GI-3-S1-EC-RDM
ECTS : 3

## HOURS

| | |
|---|---|
| Cours : | 0h |
| TD : | 30h |
| TP : | 2h |
| Projet : | 0h |
| Evaluation : | 2h |
| Face à face pédagogique : | 34h |
| Travail personnel : | 12h |
| Total : | 46h |

## ASSESMENT METHOD

EE (évaluation écrite: poster, rapports) : mini-project coefficient 1
IE (interrogation écrite) : written test 1h50 coefficient 2

## TEACHING AIDS

Duplicated course material

## TEACHING LANGUAGE

French

## CONTACT

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## AIMS

This course belongs to teaching unit Conception de produits et systèmes industriels (GI-3-S1-UE-CPSI) and contributes to the following skills :

C2 Modeling and designing an information, decision and production system, of goods and services (Level 1)
C3 Evaluating, prototyping and simulating a system (Level 1)
C4 Sizing the hardware and / or software of a system (Level 1)

## CONTENT

Classes and directed works : Strength Of Materials :
* Stress modelling under simple load
* Static balance and torque of mechanical actions
* Stress tensor, strain tensor and strength criteria
* Constitutive laws (Hooke's laws)
* Bending deflection
* Introduction to the Finite Element method
* Application to a mini-project

- To Acquire the fundamental concepts of Strength Of Materials in order to analyze the behavior of mechanical components.

## BIBLIOGRAPHY

S. TIMOSHENKO, Résistance Des Matériaux, Librairie Polytechnique Béranger, 1963.
P. AGATI, F. LEROUGE et M. ROSSETTO, Résistance des Matériaux, Dunod, 1999.
D. GAY et J. GAMBELIN, Dimensionnement des Structures, Hermes, 1999.

## PRE-REQUISITES

* Mechanic : static balance
* Mathematics : integral and differential calculation bases and matrix algebra